

Optimizing Next-Day Rainfall Predictions in Australia Using Machine Learning

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Introduction

Precipitation forecasts traditionally rely on Numerical Weather Prediction (NWP) models. Recently, machine learning has been introduced to improve these predictions.

ML workflows involve many preprocessing choices such as imputation and feature engineering however it's unclear whether these decisions meaningfully impact performance.¹

Research Question: How much do preprocessing decisions actually affect next-day rainfall prediction accuracy?

Dataset

- **Source:** Australian Bureau of Meteorology
- **Scale:** ~145,000 daily observations across 30+ locations over 10 years
- **Features:** 23 including Prior-day Rainfall, Humidity, Pressure, Wind Gust Speed, etc.
- **Target:** *Rain Tomorrow* (Yes/No) ²
- **Challenge:** Class imbalance (~20% rainy days), many missing values, several correlated features

Imputation Methods

Missing values were handled using four approaches:

- **Removal:** Eliminated rows with missing values, resulting in significant data loss (>60%)
- **Mean/Mode:** Simple baseline imputation using feature averages
- **Predictive:** Trained Random Forest models to estimate missing values using other features
- **Hybrid:** Combined predictive and mean/mode imputation based on model accuracy

Feature Engineering

- Removed highly correlated features (e.g., Temp 9am and min Temp) to reduce redundancy.

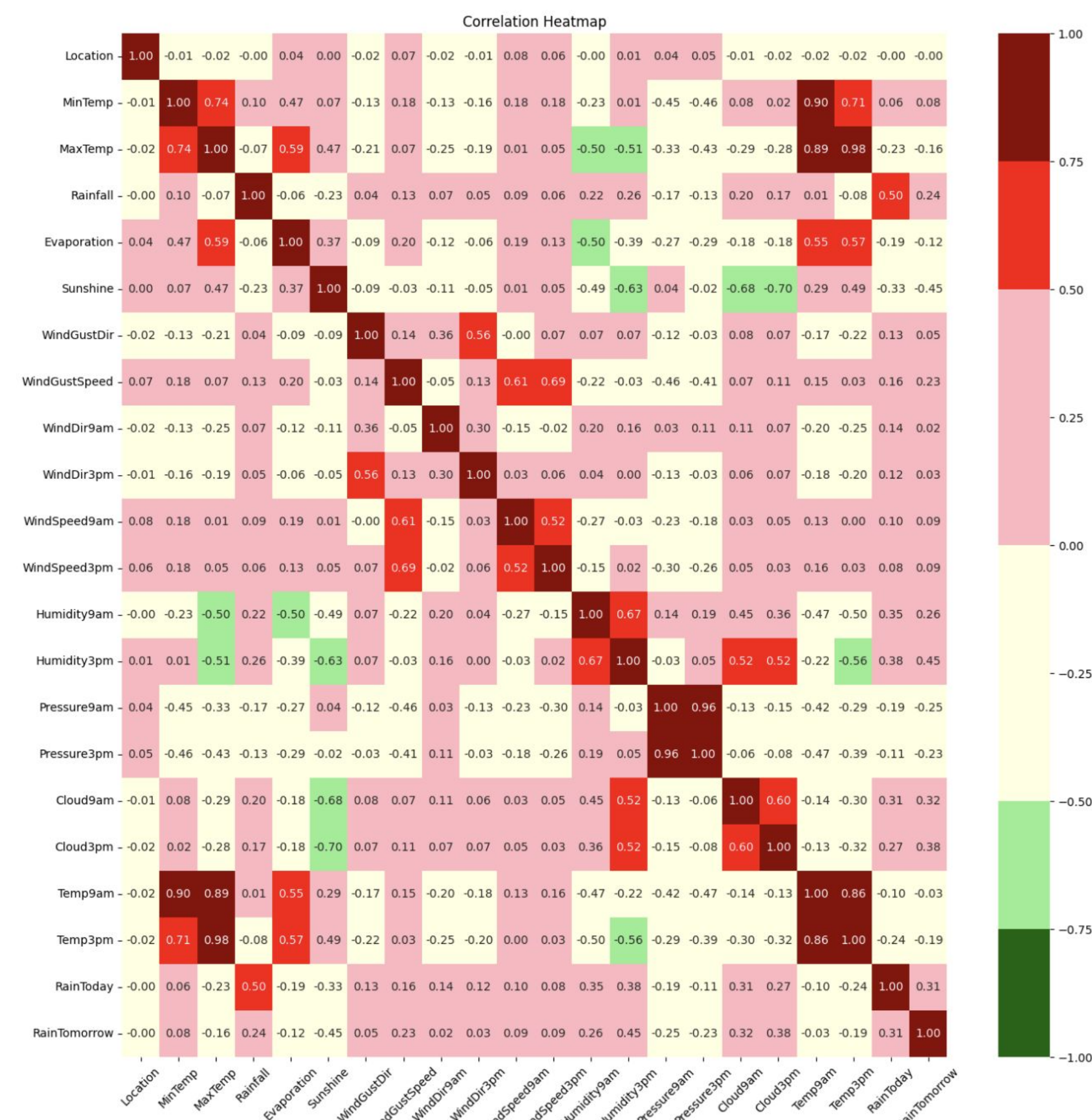


Figure 1. Correlation Matrix for all features in the dataset. Dark red boxes signify a higher correlation between features.

Experiment Setup

- **Feature engineering variations (20+)**
- **Imputation methods (4)**
- **Base Learners (4):** KNN, Decision Tree, Random Forest, Linear Classifier with SGD
- **Performance metrics (5):** Accuracy, No precision, No recall, Yes precision, Yes recall

Results - Imputation

Imputation Method	Mean Accuracy (%)
Removal	85.08
Hybrid	84.21
Regression/Classification	84.13
Mean/Mode	84.12

Figure 2. Comparison of mean classification accuracy across different imputation strategies.

Removal of missing values yields the highest accuracy likely because the dataset is large enough to tolerate data loss

Results - Feature Engineering

Feature Variations	Mean Accuracy (%)
rm-1	84.63
rm-temp2	84.62
rm-temp1	84.61
rm-pres2	84.59

Figure 3. Mean accuracy across feature ablation experiments.

Feature removal definitions:

- rm-1: Temp9am, MaxTemp removed
- rm-temp2: MinTemp, MaxTemp removed
- rm-temp1: Temp9am, Temp3pm removed
- rm-pres2: Pressure3pm removed

Removing individual temperature or pressure variables results in minimal changes in performance, indicating robustness to feature variation.

Results - Base Learners

Base Learner Models	Mean Accuracy (%)
Random Forest	85.30
SGD	84.25
KNN	84.14
Decision Tree	83.84

Figure 4. Comparison of mean classification accuracy across base learner models.

Random Forest achieves the highest performance.

Results - Overall Performance

Performance Metric	Min	Mean	Max
Accuracy	0.813540	0.843823	0.853430
No precision	0.848500	0.862795	0.874305
No recall	0.915751	0.949809	0.956826
Yes precision	0.693918	0.737016	0.753249
Yes recall	0.411174	0.470720	0.517099

Figure 5: This table above shows the Mean, minimum, and maximum score for each performance metric across all imputation methods and feature engineering combinations, averaged over all base learners

Results:

Although overall accuracy remains high and stable (81-85%), low recall for the rain class (Yes recall) limits the model's ability to detect rainy events.

Conclusion

- Model performance measured in accuracy was **stable across preprocessing choices (81%-85%)**
- Feature selection and imputation choices had **minimal practical impact** on prediction accuracy.
- **Random Forest** achieved the best overall performance among the base learners tested.
- The model is biased toward predicting *no rain*, reflecting **class imbalance**.

References

¹Price, I., Sanchez-Gonzalez, A., Alet, F., et al. (2025). *Probabilistic weather forecasting with machine learning*. **Nature**, 637, 84–90. <https://doi.org/10.1038/s41586-024-08252-9>

²Bureau of Meteorology. (2010). Rain in Australia [Data set]. Kaggle. <https://www.kaggle.com/datasets/joapauloborba/rain-in-australia>